

PrivateFin

A Privacy-First Explainable Financial Advisory System

Topic 4 — Project Design Document

Template: CM3020 Artificial Intelligence — Project Idea 2: Financial Advisor Bot
University of London | Final Year Computer Science Project

1. Project Overview

PrivateFin is a locally-executed, explainable AI financial advisory application designed for general and novice retail investors. The system generates structured, metric-anchored investment recommendations using Meta's Llama 3 large language model, served entirely on-device via Ollama on Apple M1 hardware. By eliminating cloud-side inference, PrivateFin guarantees that no personal financial query data is transmitted to external servers — a property termed **Zero Data Egress**. Recommendations are accompanied by a mandatory Chain-of-Thought (CoT) reasoning trace, ensuring every output is auditable, explainable, and compliant with the EU AI Act (2024) and GDPR Articles 5, 22, and 25.

The system is positioned to close the **Transparency-Privacy Gap** identified in the literature review: existing high-performance financial AI systems (BloombergGPT, FinGPT) are either proprietary and cloud-dependent, or open-source but still reliant on cloud APIs at inference time. PrivateFin is the first system to integrate local LLM inference, deterministic quantitative grounding, and structured CoT explainability in a single privacy-preserving architecture.

2. Domain and Users

Domain: Retail financial advisory and investment analysis.

The primary user groups are **general retail investors** and **novice investors** — individuals who wish to make informed investment decisions without access to professional financial advisors or expensive proprietary platforms. Research by the FCA (2023) found that 68% of retail investors are unaware their interaction data with financial platforms is used for commercial model training, indicating a critical and unmet need for privacy-preserving tools in this space.

Key user needs derived from the literature:

- **Privacy:** Users must be confident their financial queries and portfolio interests are not transmitted to third-party servers (GDPR Art. 25; FCA, 2023).
- **Explainability:** Novice investors require human-readable justifications for recommendations, not opaque outputs. The EU AI Act (2024) Art. 13 mandates this for high-risk AI systems.

- **Accessibility:** The tool must run on consumer hardware (M1 Mac) without cloud subscriptions, addressing the access gatekeeping problem of systems like BloombergGPT.
- **Trustworthiness:** Recommendations must be grounded in verifiable, deterministic market data (RSI, MACD) rather than unverifiable social sentiment (cf. FinGPT's Twitter/Reddit pipeline).

3. System Architecture and Design

PrivateFin follows a four-layer pipeline architecture. Each layer is designed to be independently testable, ensuring that failures can be isolated and the system's audit trail is preserved at every stage.

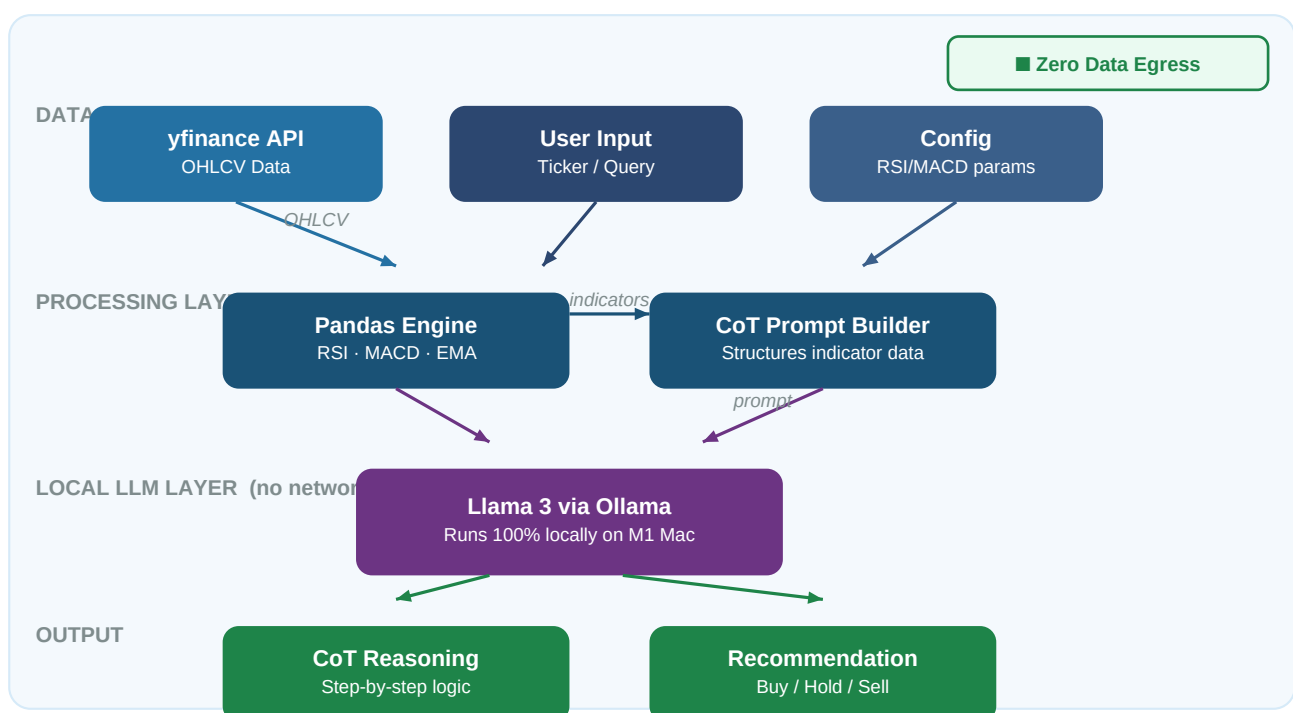


Figure 1: PrivateFin System Architecture — full local execution, zero network data egress.

Layer 1 — Data Ingestion: The yfinance library retrieves OHLCV (Open, High, Low, Close, Volume) time-series data for a user-specified ticker. No API key is required, eliminating user-linked data trails on third-party servers.

Layer 2 — Processing: A Pandas engine computes deterministic technical indicators: RSI (14-period, Wilder 1978) and MACD (EMA12 – EMA26, signal EMA9, Appel 1979). These closed-form calculations produce identical outputs for identical inputs, forming the verifiable audit trail.

Layer 3 — CoT Prompt Builder: A structured prompt template injects the computed indicator values, their conventional thresholds, and explicit reasoning instructions into a prompt sent to Llama 3. The template enforces a logical reasoning scaffold: *Observe* → *Interpret* → *Infer* → *Recommend*.

Layer 4 — Local LLM (Llama 3 via Ollama): The model generates a CoT reasoning trace followed by a Buy / Hold / Sell recommendation. All inference occurs on-device; no data leaves the machine.

4. Key Technologies and Methods

Technology	Role	Justification
Llama 3 (Meta)	Core reasoning LLM	Open-source, quantisable to 4-bit for M1; capable of instruction-following and CoT generation
Ollama	Local model server	Enables fully local inference with no cloud dependency; simple Python API
yfinance	Market data ingestion	No API key required; deterministic, reproducible OHLCV data for any ticker
Pandas	Technical indicator computation	Industry-standard; RSI and MACD calculations are closed-form and auditable
Python 3.11	Application runtime	Ecosystem compatibility with all libraries; native M1 support
Streamlit / CLI	User interface	Lightweight local UI; no server required; accessible to non-technical users

5. Work Plan

The project spans Weeks 1–22, structured across four topics. The Gantt chart below maps all major tasks, with contingency periods shown as dashed bars. Tasks are sequenced to ensure evaluation is embedded throughout rather than deferred to the final phase.

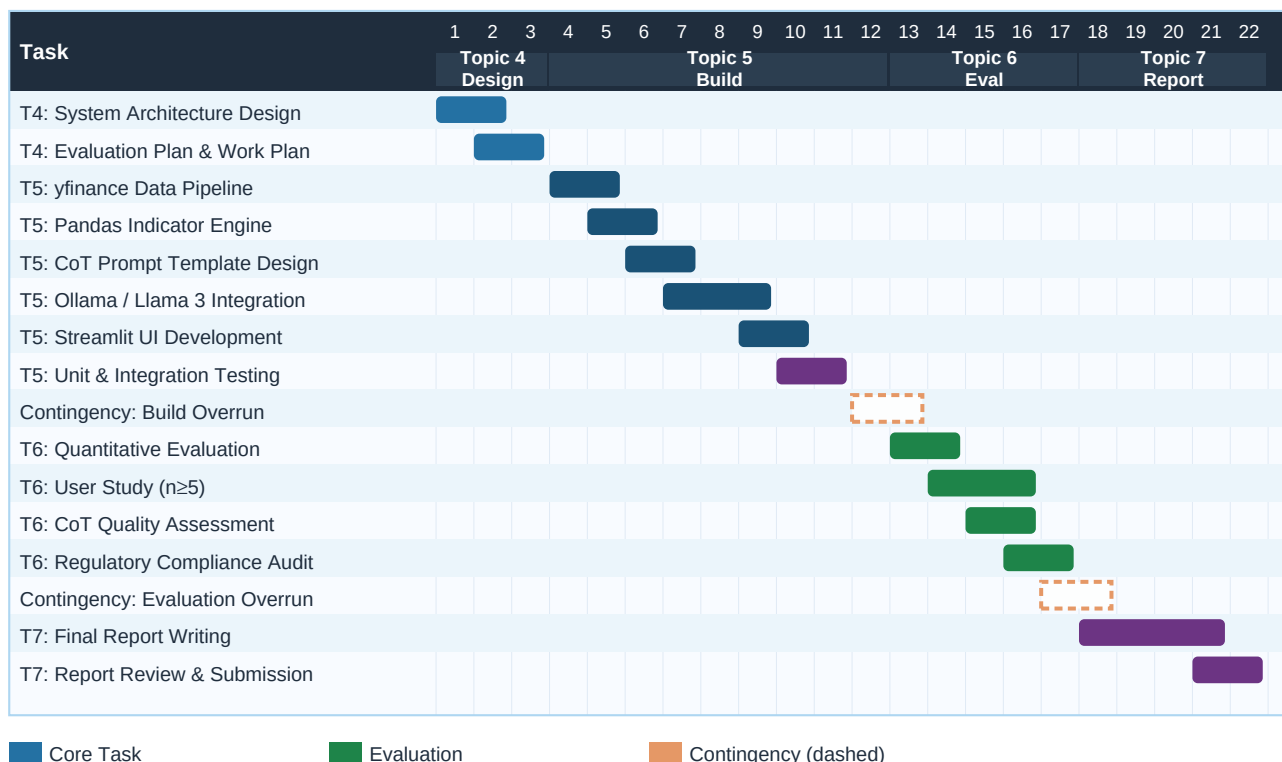


Figure 2: Project Gantt Chart (Weeks 1–22). Dashed bars indicate contingency periods.

Feasibility Assessment:

The core technical stack (yfinance, Pandas, Ollama) is mature and well-documented, reducing implementation risk. Llama 3 7B quantised to 4-bit runs at approximately 15–20 tokens/second on M1 hardware, which is sufficient for interactive advisory use. Two explicit contingency buffers (Weeks 12–13 and Weeks 17–18) are allocated to absorb integration delays or model performance issues. The evaluation phase begins at Week 13, ensuring sufficient time for a rigorous multi-method assessment before the Week 22 submission deadline.

Contingency Plans:

- **Risk: Llama 3 CoT quality insufficient.** Mitigation: Switch to prompt chaining (decompose reasoning into multiple sequential prompts). Two weeks of contingency allocated at Week 12.
- **Risk: yfinance API deprecation or data gaps.** Mitigation: Fall back to Alpha Vantage free tier (5 calls/min); data pipeline is abstracted behind an interface for easy substitution.
- **Risk: M1 hardware performance bottleneck.** Mitigation: Reduce quantisation precision from 4-bit to 2-bit; limit context window length.
- **Risk: User study recruitment shortfall.** Mitigation: Minimum viable n=5; supplement with expert evaluation (self-evaluation against regulatory explainability rubric).

6. Evaluation Plan

The evaluation is designed to assess the three core claims of PrivateFin: (1) that it preserves user privacy, (2) that its recommendations are grounded in correct deterministic calculations, and (3) that its CoT reasoning constitutes meaningful explainability in the sense required by the EU AI Act (2024). A multi-method approach is used, combining automated quantitative testing, regulatory compliance auditing, and a structured user study.

Evaluation Component 1: Quantitative Pipeline Correctness

Method: Unit testing against known-correct values. RSI and MACD values computed by PrivateFin will be validated against outputs from a reference implementation (TA-Lib) on 10 historical tickers across 3 time periods each. **Success criterion:** 100% agreement to 4 decimal places (deterministic calculation — any deviation indicates a bug).

Evaluation Component 2: Privacy Architecture Audit

Method: Network traffic analysis using Wireshark during a live session. All outbound connections will be logged during a 10-query evaluation session. **Success criterion:** Zero data transmitted to any endpoint other than Yahoo Finance (for OHLCV data retrieval). This provides verifiable evidence of the Zero Data Egress claim.

Evaluation Component 3: CoT Explainability Quality

Method: A structured rubric-based assessment of 20 generated recommendations across 10 tickers (2 per ticker at different market conditions). Each CoT output will be scored on four dimensions:

- **Factual Accuracy:** Do the indicator values cited in the reasoning match the computed values? (0–2)
- **Logical Coherence:** Does the inference follow from the cited data? (0–2)
- **Completeness:** Are all four CoT stages present (Observe → Interpret → Infer → Recommend)? (0–2)
- **Regulatory Compliance:** Does the output satisfy EU AI Act Art. 13 transparency requirements? (0–2)

Success criterion: Mean total score $\geq 6/8$ across all 20 samples. Outputs scoring below 4/8 will be analysed qualitatively to identify systematic prompt template failures.

Evaluation Component 4: User Study

Method: A task-based user study with a minimum of 5 participants drawn from the target user groups (general and novice investors). Participants will be asked to: (a) query PrivateFin for 3 tickers of their choice; (b) rate the understandability of the CoT explanation on a 5-point Likert scale; and (c) complete the System Usability Scale (SUS, Brooke 1996). A brief semi-structured interview will capture qualitative feedback on trust and privacy perception. **Success criteria:** Mean SUS score ≥ 70 (above the 'Good' usability threshold); mean explanation understandability $\geq 4/5$.

Evaluation Summary

Component	Method	Success Criterion	Addresses Aim
Pipeline Correctness	Unit tests vs TA-Lib	100% agreement (4 d.p.)	Claim 2: Determinism
Privacy Audit	Wireshark traffic analysis	Zero unauthorised egress	Claim 1: Privacy
CoT Quality	Rubric scoring (n=20)	Mean $\geq 6/8$	Claim 3: Explainability
User Study	SUS + Likert + interview	SUS ≥ 70 , UX $\geq 4/5$	Claims 1, 3: Trust & UX

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